

Problem. Each decision step of the cached executor observes a single scalar, the fused retrieval score s_t , and must choose one of $K+1$ actions: full inference (action 0) or one of K reuse tiers $a \in \{1, \dots, K\}$ of increasing skip depth $\tau_1 < \dots < \tau_K$ (in the current system $K = 2$: warm start at $\text{start_t} = 0.3$ and full replay). Unit compute costs are fixed and known,

$$c_a = s_1 + \mathbf{1}[a < K] (s_2 + \text{start_t}(a) \cdot s_3), \quad \text{start_t}(0) = 1, \quad \text{start_t}(K) = 0, \quad (1)$$

so that $c_K < \dots < c_1 < c_0 = s_1 + s_2 + s_3$, with s_1, s_2, s_3 the measured stage latencies (vision encoding, language-model prefix, full flow integration). Every decision pays s_1 , which produces the features the retrieval score is computed from; a warm tier resumes the flow at $\text{start_t}(a)$ and pays that fraction of s_3 ; the deepest tier replays the cached chunk and skips both the prefix and the integration. The online rule class we consider is the incumbent one: *monotone cut vectors*. A rule is a vector $\theta = (\theta_1, \dots, \theta_K)$, $\theta_1 \leq \dots \leq \theta_K$ interpreted through “execute the deepest tier whose cut is cleared”, $a(s) = \max(\{0\} \cup \{a : s \geq \theta_a\})$. Every policy discussed in this note—the hand-tuned threshold dispatch, its percentile-indexed grid, and the calibrated rule—is a member of this class at run time; a deployment differs from the incumbent only in *how θ is chosen offline*. This note makes that difference the object of study.

Two indexings of one family. The incumbent *rate indexing* sets each θ_a at a percentile of the development score distribution: $\theta_a = \hat{F}_s^{-1}(1 - f_a)$ admits a target fraction f_a of decisions into tier a or deeper. Rates are predictable by construction; *risk* is not. Which vector $(f_1, \dots, f_K)$ lands on the cost–success frontier is unknown a priori, so the incumbent finds operating points by rolling out a K -dimensional grid. The alternative studied here is *risk indexing*. Let $Y_a \geq 0$ be the offline-checkable deviation surrogate of the companion note (Eq. 2 there) for tier a , and define per tier the conditional upper quantile

$$q_a(s) = Q_{1-\alpha}(Y_a \mid s_t = s), \quad a = 1, \dots, K, \quad (2)$$

estimated on calibration rollouts by order-restricted (isotonic) quantile regression under the monotonicity assumption (A1’): q_a nonincreasing in s , nondecreasing in the skip depth. Given one tolerance δ , the cuts are *computed*, not searched:

$$s_a^*(\delta) = \min\{s : \hat{q}_a(s) \leq \delta\}, \quad \theta_a = s_a^*(\delta). \quad (3)$$

The rule executes the deepest admissible tier and falls back to full inference when no tier is admissible (fail-closed, as in the companion note). Online, Eq. 3 is indistinguishable in form and cost from the incumbent: the same comparisons against stored constants.

Equivalence at $K = 1$ (a concession). With a single tier the two indexings coincide as families. Under (A1’) the map $\delta \mapsto s_1^*(\delta)$ is a monotone reparametrization of the cut, so $\{\text{risk-indexed rules}\}_\delta = \{\text{threshold rules}\}_\theta = \{\text{monotone rules on } s\}$, and under the monotone-likelihood-ratio premise already stated in the companion note the common family traces the ROC frontier of s (Neyman–Pearson). At $K = 1$, therefore, risk indexing buys semantics but no new operating points, and percentile inversion is equally automatic. Any claimed contribution must—and here does—begin at $K \geq 2$.

Coupling at $K \geq 2$: the solution path. Consider the budgeted design problem over cut vectors,

$$\max_{\theta} \text{SR}(\theta) \quad \text{s.t.} \quad \mathbb{E}[c_{a(s;\theta)}] \leq B. \quad (4)$$

Introduce the surrogate objective in which closed-loop degradation is exchanged for accumulated per-decision surrogate risk,

$$(A2) \quad \text{SR}(\theta) \approx \Phi\left(\mathbb{E}[r_{a(s;\theta)}(s)]\right), \quad r_a(s) = \text{a nondecreasing functional of } q_a(s), \quad r_0 \equiv 0, \quad (5)$$

with Φ decreasing. (A2) is an approximation, not a theorem: it ignores sequencing and closed-loop intervention (the composition failure of the companion note) and is adopted because its consequences are checkable. Under (A1’), (A2), and $c_K < \dots < c_0$, the Lagrangian of Eq. 4 separates across decisions and across adjacent tier boundaries: at an optimum, each boundary θ_a equalizes marginal surrogate risk against marginal compute saving with the *same* multiplier λ ,

$$\frac{\partial \mathbb{E}[r]}{\partial \theta_a} = \lambda \frac{\partial \mathbb{E}[c]}{\partial \theta_a} \quad (a = 1, \dots, K), \quad (6)$$

so the set of optimal cut vectors is a *one-parameter solution path* indexed by λ . When the marginal risk at a boundary is governed by the calibrated quantile at the cut—the case in which admission at θ_a adds decisions of surrogate risk $\approx \hat{q}_a(\theta_a)$ —the path is traced by a common level: $\hat{q}_a(\theta_a) = \delta(\lambda)$ for all a , which is exactly Eq. 3. The calibrated ladder $\{\theta(\delta) : \delta \geq 0\}$ is thus the plug-in estimate of the solution path of the K -dimensional design problem, and a K -dimensional grid search rediscovers that path pointwise. Two consequences are testable: (i) ladder points should track the grid family’s empirical Pareto hull at matched evaluation budget; (ii) at a fixed number n of a-priori arms, the vertical regret of a K -dimensional grid against its own hull grows with K , while the ladder remains one-dimensional. Neither consequence asserts that the path *dominates* the grid family—it cannot, being a subset; the claim is that the path attains the family’s frontier without the search.

Per-task and cross-library transfer. Rate indexing conditions on nothing: a percentile vector tuned on one task mixture or one library encodes that mixture’s score distribution, and the incumbent’s own documentation records that carrying a threshold across libraries is a known failure mode. Risk indexing recomputes Eq. 3 from the calibrated curves: per-task (Mondrian) cuts $s_{a,t}^*(\delta)$ follow from per-task quantile curves at the same δ when sample sizes permit, and a library change re-runs calibration rather than a K -dimensional (per task, $K \times T$ -dimensional) grid. The empirical finding that task-conditional selection improves every family by a comparable margin is consistent with this reading: conditioning is a property of the information set, not of any single rule family; what risk indexing changes is the *cost of exploiting it*.

What is not claimed. No statement here is a closed-loop guarantee: deployment intervenes on the visitation distribution, and Eq. 5 is a surrogate adopted for its checkable consequences (the companion note’s scope paragraph applies verbatim). The path property is relative to rules measurable in s alone; richer information sets—the disagreement statistic v , task identity used online, stateful gates—can move the frontier, and whether they do is an empirical question that our measurements answer in the negative for v (no measurable increment at matched density) and in the positive, equally for all families, for task identity. Finally, the estimated path can miss the frontier where calibration data are sparse; matched-density evaluation, not construction, is what supports the tracking claim.

Experimental consequences. Four readouts follow. (i) *Matched-density frontier comparison*: the risk ladder against the dense percentile grid on the same evaluation set, library, and cost accounting; the tracking claim predicts hull agreement within estimation noise. (ii) *Tuning budget*: frontier quality (hull area or vertical regret) as a function of the number of a-priori arms for each indexing; the prediction is a widening gap as arms are removed, and a qualitative jump when K grows. (iii) *Risk calibration*: realized per-decision surrogate risk of admitted steps against the declared δ —the plot only a risk-indexed rule can draw in advance. (iv) *System composition*: the full deployed stack adds the production hysteresis gate in front of either indexing; the ceiling-equality reading predicts that gate plus risk ladder matches gate plus tuned grid at matched freedom, measured on the same chain. A three-tier instance ($K = 3$, second warm depth unpinned from `start.t` = 0.3 after its unit cost is frozen) turns consequence (ii) into the discriminating experiment: budget-matched 3-dimensional grid against the unchanged one-dimensional ladder.